

# Graphical Program Slicing with Galois Dependence Graphs

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# Background

# Fluid

- Call-by-value dynamic functional calculus
  - Surface and core languages
  - Records and algebraic datatypes
- Bi-directional dependency Analysis
  - Make selections on output and compute the necessary input
  - Selection information can be followed through both evaluation and desugaring stage
  - Useful for visualization tasks especially in - for example - data science



# Dynamic Program Slicing

- Use dynamic dependence graphs
  - Introduced in 1990<sup>(1)</sup>
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(1) Hiralal Agrawal and Joseph R. Horgan. 1990. Dynamic program slicing. SIGPLAN Not. 25, 6 (Jun. 1990), 246–256. <https://doi.org/10.1145/93548.93576>



Next: Graph Semantics